



TECHNICAL SPECIFICATION

GPAY PAYMENT SERVICES

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Table of Content

1. Introduction	3
2. Technical specification	3
2.1 Environment	3
2.1.1 API Endpoint	3
2.1.2 Swaggerhub	3
2.1.3 Merchant Authentication	3
2.1.4 Public Key	3
2.1.5 Data test	3
2.2 Authentication and Security	4
2.2.1 Digital Signature	4
2.3 APIs specification	5
2.3.1 Query bank account information	5
2.3.2 Fund transfer wallet to bank account	6
2.3.3 Get transfer detail by merchant transaction id	7
2.3.4 Get merchant account balance	9
2.4 Response Code	10

1. Introduction

GPAY payment services allows Merchant to create payment order to pay from Merchant's e-wallet balance to other e-wallet account or bank account.

The document contains API specification, interface, message, data format and message format.

APIs in this document:

- *Fund transfer wallet to wallet*
- *Fund transfer wallet to Bank Account*
- *Get Transaction Detail*
- *Get merchant Account balance.*

Data fields in this document:

- Field Name
- Field Type
- Field required or not.
- Max length: Maximum length of field
- Field description

2. Technical specification

2.1 Environment

2.1.1 API Endpoint:

- Sandbox environment: <https://mpa.sandbox.g-pay.vn/api/v3>
- Production environment: <https://mpa.g-pay.vn/api/v3/>

2.1.2 Swaggerhub: [Link](#)

2.1.3 Merchant Authentication:

- merchant code: will be issued privately to merchant
- password: will be issued privately to merchant

2.1.4 Public Key

- public key for verify signature:

2.1.5 Data test

Account test:

No	Type	Account Number	Full name	Bank Code	Note
1	Account number	0687041020972	NGUYEN THI XUAN	VCCB	
2	Card number	9704542000234196	NGUYEN THI XUAN		With successful case, please enter the above card information and the message field =Test200

2.2 Authentication and Security

2.2.1 Digital Signature

To ensure that the data of both parties is not tampered with, the two parties will use digital signatures on the exchanged data. The rule for generating digital signatures (the signature field) is made according to the following formula:

```
openssl_sign($input, $binary_signature, $privateKey,  
OPENSSL_ALGO_SHA256); $signature = base64_encode($binary_signature);
```

INPUT field structure:

INPUT: Field_Name1=Field_Value1& Field_Name2=Field_Value2.....

List Fields to generate signature depend each API

Example INPUT: merchant_code={} & order_id={} & amount={}.

- *merchant_code: Merchant code*
- *order_id: Order ID*
- *amount: amount of order*

INPUT:

merchant_code=MERCHANT_XYZ&transaction_id=12345678&amount=10000

2.2.2 SSL

Connection between Merchant and GPAY uses HTTPS connection protocol with TLS1.2 protocol; All data will be securely encrypted during data transmission.

2.3 APIs specification

2.3.1 Query bank account information

URL	/fund-transfers/inquiry
HTTP Method	POST
Authentication	• Digital Signature. • SSL
Protocol	HTTPS/REST-JSON

No	Fields	Data type	Required	Length	Description
API Input					
1	request_id	String	Y	50	The transaction id code is unique generated by the merchant
2	merchant_code	String	Y	50	Merchant code
3	account_number	String	Y	50	Account number/card number to be queried
4	type	String	Y	50	The type of account_number: [ACCOUNT_NUMBER, CARD_NUMBER]
5	bank_code	String	N	50	List of bank codes: Click here Query card number information without transmitting bank_code
6	signature	String	Y	2048	The INPUT has the following structure: <i>merchant_code={}&account_number={}&type={}&request_id={}</i> Example: merchant_code=MERCHANT_XYZ&account_number=0123456789&type=ACCOUNT_NUMBER&request_id=MT00011123
API Output					
1	meta.code	string	Y	10	API processing response status
2	meta.msg	string	Y	256	Response message
3	meta.internal_msg	string	Y	256	Detail response message
4	response.order_id	string	Y	50	Query transaction code generated by GPAY
5	response.status	string	Y	20	The status of transfer: [ORDER_PENDING, ORDER_PROCESSING, ORDER_SUCCESS, ORDER_FAILED, ORDER_CANCEL, ORDER_VERIFYING]

6	response.full_name	string	Y	255	Cardholder/Account holder name
7	response.signature	string	Y	2048	The data has the following structure: order_id={} &full_name={} &status={} transfer_id: The id of transfer order_id: order id transaction GPAY transfer_status: The status of transfer Example: order_id=GPOS1234567890&full_name=NGUYEN VAN A&transfer_status=ORDER_SUCCESS

2.3.2 Fund transfer wallet to bank account/card number

URL	/fund-transfers/ft-to-bank
HTTP Method	POST
Authentication	• Digital Signature • SSL
Protocol	HTTPS/REST-JSON

No	Fields	Data type	Required	Length	Description
API Input					
1	account_number	String	N	20	Bank account number
2	bank_code	String	N	20	The bank code(required if type is ACCOUNT_NUMBER) Link
3	full_name	String	N	200	Full name of account holder
4	amount	int	Y		The amount to transfer
5	merchant_code	String	Y	50	The code of merchant
6	transaction_id	String	Y	50	The transaction code is unique generated by the merchant
7	type	String	N	20	The type of account_number: [ACCOUNT_NUMBER, CARD_NUMBER]
8	order_ref	String	N	50	The reference code of the inquiry transaction

					returned by Gpay(order_id in response). Merchant can fill in or not. If Merchant does not fill in, GPAY will make inquiry before making the money transfer. If enter order_ref, may not need to enter the information account_number, bank_code, type
9	map_id	String	N	50	Reference code for disbursement transaction. Merchant can use it to add a reference code such as: contract number, recipient's phone number,....
10	message	String	N	200	Transfer content
11	signature	String	N	50	The input has the following structure: merchant_code={} &account_number={} &bank_code={} &order_ref={} &amount={} &transaction_id={} &type={} <i>Example:</i> merchant_code=MERCHANT_XYZ&account_number=0687041020972&bank_code=VCCB&order_ref=OS12345676&amount=10000&transaction_id=MT2009&type=ACCOUNT_NUMBER

API Output

1	meta.code	string	Y	10	API processing response status
2	meta.msg	string	Y	256	Response message
3	meta.internal_msg	string	Y	256	Detail response message
4	response.transfer_id	string	Y	50	The id of transfer
5	response.transfer_status	string	Y	20	The status of transfer: [ORDER_PENDING, ORDER_PROCESSING, ORDER_SUCCESS, ORDER_FAILED, ORDER_CANCEL, ORDER_VERIFYING]
6	response.order_id	string	Y	50	order id transaction of GPAY
7	response.transfer_created_time	String	Y	20	The time that created transfer object
8	response.transf	String	Y	20	The last time that update status

	e r_status_updated_time				
9	response.signal	string	Y	2048	The data has the following structure: transfer_id={} & order_id={} & transfer_status={} transfer_id: mã giao dịch chuyển tiền order_id: mã giao dịch của Gpay transfer_status: Trạng thái giao dịch chuyển tiền Example: transfer_id=202003040000000014 & order_id=GPOS1234 567890 & transfer_status=ORDER_SUCCESS

2.3.3 Get transfer detail by merchant transaction id

URL	/fund-transfers/{w2b}/{fund_transfer_id}
HTTP Method	POST
Authentication	• Digital Signature • SSL
Protocol	HTTPS/REST-JSON

No	Fields	Data type	Required	Length	Description
Parameter					
	fund_transfer_id	String	Y	2	Merchant's transfer transaction id
	w2b	String	Y	2	type of transfer (wallet to wallet bank account); Available values : w2w, w2b
API Input					
1	merchant_code	String	N	256	The code of merchant

2	signature	String	Y	2048	The input has the following structure: <code>merchant_code={} & id={} </code> merchant_code: Code of merchant id: Merchant transfer transaction id Example: <code>merchant_code=MERCHANT_XYZ&id=MT00011</code>
API Output					
1	meta.code	string	Y	10	API processing response status
2	meta.msg	string	Y	256	Response message
3	meta.internal_msg	string	Y	256	Detail response message
4	response.transfer_id	string	Y	50	The transfer id
5	response.transfer_status	string	Y	20	The status of transfer: [ORDER_PENDING, ORDER_PROCESSING, ORDER_SUCCESS, ORDER_FAILED, ORDER_CANCEL, ORDER_VERIFYING]
6	response.order_id	string	Y	50	order id transaction of GPAY
7	response.transfer_created_time	String	Y	20	The time that created transfer object
8	response.transfer_status_updated_time	String	Y	20	The last time that update status
9	response.signature	string	Y	2048	The data has the following structure: <code>transfer_id={} & order_id={} & transfer_status={} </code> transfer_id: The id of transfer order_id: order id transaction GPAY transfer_status: The status of transfer Example: <code>transfer_id=GPOS1234&order_id=GPOS1234567890&transfer_status=ORDER_SUCCESS</code>

2.3.4 Get merchant account balance

URL	/merchants/get-merchant-account-information
HTTP Method	POST
Authentication	• Digital Signature • SSL
Protocol	HTTPS/REST-JSON

No	Fields	Data type	Required	Length	Description
API Input					
1	merchant_code	String	N	256	The code of merchant
2	signature	String	Y	2048	The input has the following structure: "merchant_code={}". Example: merchant_code=MERCHANT_XYZ"
API Output					
1	meta.code	string	Y	10	API processing response status
2	meta.msg	string	Y	256	Response message
3	meta.internal_msg	string	Y	256	Detail response message
4	response.amount_revenue	int	Y		Revenue amount
5	response.amount_cash	int	Y		Available payment amount
6	response.amount_minimum	int	Y		Minimum amount
7	response.amount_refund	int	Y		Refund amount
8	response.cash_in_information.account_number	string	Y		Account number for cashin

9	response.cash_in_information.account_name	string	N		Account name for cashin
10	response.cash_in_information.bank_code	string	N		Bank code for cashin
11	response.cash_in_information.bank_name	string	N		Bank Name for cashin
12	response.cash_in_information.qr_code	string	N		vietQR string(use to scan QR on the app banking has support vietQR)
13	response.cash_in_information.qr_code_image	string	N		VietQR image(data type base64)

2.4 Response Code

API processing status is returned via **the meta.code** field..

meta.code = 200 => API process success

meta.code # 200 => API process error

meta.code = 202 => Transaction timeout/verifying

Error Codes (attached with this document)